

# MACHINE LEARNING FUNDAMENTALS SERIES

## The Essential Main Ideas of Neural Networks

### Inside the Black Box: How Neural Networks Transform Activation Curves to Fit Complex Squiggles

Source: StatQuest with Josh Starmer — The Essential Main Ideas of Neural Networks

**Simplified Demonstration Architecture & Pre-trained Weights:** The neural network presented in this document is a *minimalist 1-hidden-layer network with 2 hidden nodes*, designed specifically for step-by-step conceptual illustration.

All weight values (e.g.,  $-34.4, -1.3, -2.52, 2.28$ ) and bias values (e.g.,  $+2.14, +1.29, -0.58$ ) shown throughout this example have **already been trained and optimized** using Backpropagation. The sole purpose of this lesson is to provide an intuitive, visual understanding of **how a neural network uses its parameters to slice, flip, scale, and add simple activation curves together** to "draw" or fit a non-linear curve (a "squiggle") through data.

## 1. Introduction and Problem Formulation

Neural networks are often considered "black boxes" due to their apparent complexity and dense web of connections. However, at their core, neural networks are fundamentally **"big fancy squiggle-fitting machines."**

### 1.1. The Drug Dosage Effectiveness Dataset

Consider a clinical trial testing a new drug across three patient groups given different dosages:

- **Low Dosage (0.0):** Ineffective  $\implies$  Effectiveness = 0.0.
- **Medium Dosage (0.5):** Highly Effective  $\implies$  Effectiveness = 1.0.
- **High Dosage (1.0):** Ineffective / Toxic  $\implies$  Effectiveness = 0.0.

**The Failure of Linear Models:** A standard linear regression or logistic regression model fits a straight line ( $\hat{y} = \beta_0 + \beta_1 x$ ). Regardless of how a straight line is rotated, it can at most accurately predict **two out of the three** dosage points. It is incapable of bending to capture the rise-and-fall pattern of the medium-effective dosage.

A **Neural Network** solves this problem by fitting a non-linear curve (a "green squiggle") that passes close to 0 at Low Dosage, climbs to 1 at Medium Dosage, and drops back to 0 at High Dosage.

## 2. Neural Network Architecture and Terminology

A neural network consists of **Nodes** (neurons) connected by directional **Connections** (synapses).

### 2.1. Core Terminology

- **Input Layer:** The entry node where feature values ( $x = \text{Dosage}$ ) are fed into the network.
- **Hidden Layer:** The intermediate layer(s) of nodes between Input and Output (called "hidden" because their values are not directly observed in the training dataset).
- **Weights (Multiplicative Parameters):** Numbers along connection lines that multiply incoming signals (analogous to slopes).

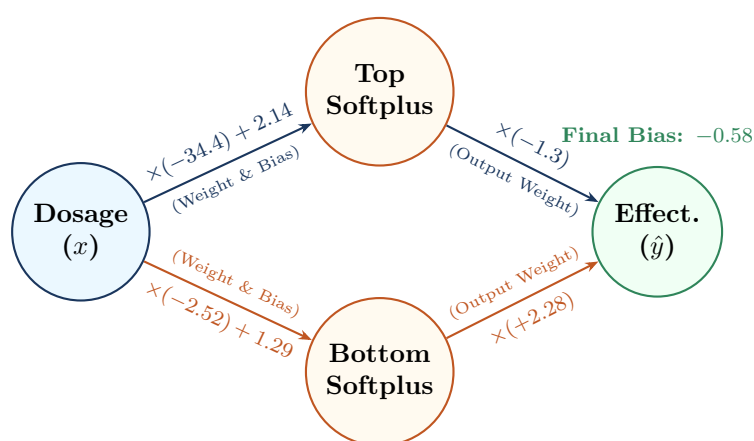


Figure 1: Architecture of the Minimalist Demonstration Neural Network (1 Input, 1 Hidden Layer with 2 Softplus Nodes, 1 Output). All connection values are pre-trained parameters.

- **Biases** (Additive Parameters): Numbers added to weighted sums before or after activation functions (analogous to y-intercepts).
- **Activation Functions:** Bent or curved functions inside hidden nodes that inject **non-linearity** into the network. Common activation functions include:
  - **Softplus:**  $y = \ln(1 + e^x)$  — Smooth, curved line (used in this StatQuest example).
  - **ReLU (Rectified Linear Unit):**  $y = \max(0, x)$  — Bent straight line (most common in modern deep learning).
  - **Sigmoid:**  $y = \frac{1}{1+e^{-x}}$  — S-shaped curve (historically popular).

### 3. Step-by-Step Mathematical Curve Transformation

The neural network constructs the final prediction curve through three sequential mathematical operations: **Slicing/Shifting**, **Weight Scaling/Flipping**, and **Summation with Final Shift**.

#### 3.1. Top Hidden Node (Blue Branch Transformation)

##### 1. Input to Hidden Transformation:

$$x_{\text{top}} = (-34.4 \cdot \text{Dosage}) + 2.14$$

For Dosage = 0:  $x_{\text{top}} = 2.14$ . For Dosage = 1:  $x_{\text{top}} = -32.26$ .

##### 2. Activation Function Evaluation:

$$y_{\text{top}} = \text{Softplus}(x_{\text{top}}) = \ln(1 + e^{x_{\text{top}}})$$

For Dosage = 0:  $y_{\text{top}} = \ln(1 + e^{2.14}) \approx 2.25$ . For Dosage = 0.1:  $y_{\text{top}} \approx 0.24$ . This maps the Dosage range  $[0, 1]$  to a narrow slice of the Softplus curve.

##### 3. Output Weight Scaling:

$$\text{Curve}_{\text{blue}} = y_{\text{top}} \cdot (-1.3)$$

Multiplying by  $-1.3$  **flips the curve upside down** and scales its magnitude, yielding a curve that starts at  $-2.93$  (when Dosage = 0) and flattens out near 0.

#### 3.2. Bottom Hidden Node (Orange Branch Transformation)

##### 1. Input to Hidden Transformation:

$$x_{\text{bottom}} = (-2.52 \cdot \text{Dosage}) + 1.29$$

For Dosage = 0:  $x_{\text{bottom}} = 1.29$ .

##### 2. Activation Function Evaluation:

$$y_{\text{bottom}} = \text{Softplus}(x_{\text{bottom}}) = \ln(1 + e^{x_{\text{bottom}}})$$

For Dosage = 0:  $y_{\text{bottom}} = \ln(1 + e^{1.29}) \approx 1.53$ .

##### 3. Output Weight Scaling:

$$\text{Curve}_{\text{orange}} = y_{\text{bottom}} \cdot (+2.28)$$

Multiplying by  $+2.28$  **stretches the curve upward**, producing an orange curve that starts high and gently slopes down as Dosage increases.

#### 3.3. Combining Curves to Form the Final Squiggle

Finally, the output node sums the scaled blue and orange curves and subtracts the final output bias ( $-0.58$ ):

$$\hat{y} = \text{Curve}_{\text{blue}} + \text{Curve}_{\text{orange}} - 0.58$$

#### 3.4. Evaluating a New Prediction

If a patient receives a dosage of 0.5 (Medium Dosage):

$$x_{\text{top}} = -34.4(0.5) + 2.14 = -15.06 \implies y_{\text{top}} = \text{Softplus}(-15.06) \approx 0.00$$

$$x_{\text{bottom}} = -2.52(0.5) + 1.29 = +0.03 \implies y_{\text{bottom}} = \text{Softplus}(0.03) \approx 0.708$$

$$\hat{y} = (-1.3 \times 0.00) + (+2.28 \times 0.708) - 0.58 = 1.614 - 0.58 = \mathbf{1.034}$$

Since  $1.034 \approx 1.0$ , the network confidently predicts that a dosage of 0.5 will be **effective**.

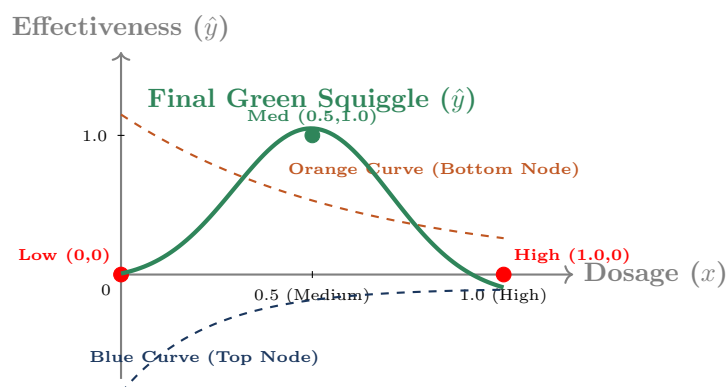


Figure 2: Visual Synthesis: The individual scaled blue and orange activation curves are added together and shifted by the final bias ( $-0.58$ ) to form the Green Squiggle fitting the 3 clinical data points.

## 4. Summary of Core Lessons

1. **Big Fancy Squiggle-Fitting Machines:** Neural networks are not mysterious black boxes; they are structured algorithms that fit flexible non-linear curves ("squiggles") to complex datasets.
2. **Role of Activation Functions:** Activation functions (Softplus, ReLU, Sigmoid) provide the essential curved or bent building blocks.
3. **Role of Weights & Biases:**
  - **Input Weights & Biases:** Slice, shift, and select specific regions of activation function curves.
  - **Output Weights:** Flip and scale the magnitude of activation curves.
  - **Output Bias:** Shifts the final combined curve up or down to align with data points.
4. **Scalability:** Adding more hidden layers and nodes allows neural networks to fit extraordinarily complex high-dimensional squiggles to virtually any dataset.